Net-Aware Learned Block Mapping for Reversible Data Hiding in Binary Images

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ARTICLE INFO

Keywords:
Binary-image reversible data hiding
PEAK/ZERO block mapping
Serialization-aware mapping selection
Auxiliary-information coding
Learned distortion ranking

ABSTRACT

This paper presents a reversible data hiding scheme for plaintext binary images under an explicit synchronization model. Its technical contribution is serialization-aware net-payload-driven activation of reversible PEAK/ZERO 3×3 mappings: the active mapping configuration is selected by useful payload after charging the exact serialized description required for deterministic recovery. The implementation represents the active PEAK subset by a combinatorial rank and the distance-one flip positions by base-9 digits; a convolutional neural network (CNN) supplies the learned distortion ranking for admissible ZERO candidates. On eight 512×512 document images, this activation raises mean net payload from 2139 to 2245 bits relative to an otherwise identical wire-charged gross-selection variant and preserves exact round-trip reversibility. In a common executable protocol on 100 disjoint thresholded BOSSbase images, the proposed adaptive block-mapping method (ABM), Dong, Huynh–Nguyen, and the opposite-center-pair method (PPOCP) are compared with identical messages, payload targets, distance-reciprocal distortion (DRD) computation, auxiliary-bit accounting, and reversibility checks. The proposed method achieves the highest useful capacity among the evaluated implementations, while PPOCP retains the lowest-distortion and most favorable detectability profile. Logistic detector AUC reaches 0.900 and 0.957 at 256 and 512 bits, placing the validated operating region in capacity-oriented reversible annotation over a lossless channel with detectability-aware payload selection.

1. Introduction

The proliferation of digital archives has increased the need to associate authentication data, annotations, and integrity evidence with images while preserving the original content. Cryptography protects message content [1], whereas steganography conceals its presence [2]. Reversible data hiding (RDH) combines these objectives with exact cover restoration, a property that is especially valuable for medical, legal, and administrative binary documents.

Binary images have two pixel states, so every modification can alter stroke continuity, topology, or local readability. Specialized RDH methods therefore rely on pattern statistics, overlapping neighborhoods, or reversible block substitutions rather than direct grayscale-domain operations [3–13]. Their practical performance depends on three coupled quantities: visual distortion, the side information required for exact synchronization, and the statistical footprint visible to a steganalyst.

Existing methods occupy complementary parts of the capacity–distortion space. Magnification-based embedding provides low distortion by increasing image size [8]; PPOCP uses overlapping opposite-center pairs for perceptually selective changes [9]; Huynh–Nguyen exploit the combinatorial capacity of 3×3 block mapping [4]; and Dong et al. adapt overlapping patterns to image statistics [7]. These contributions motivate a unified evaluation in which useful payload is measured after serialized metadata and all methods use the same messages, cover images, and distortion metric.

This work builds on established net-capacity accounting. The distinguishing algorithmic decision is that the exact serialized description cost of the reversible mapping configuration participates in deciding which PEAK/ZERO tables remain active. A convolutional neural network ranks distance-one ZERO candidates by predicted perceptual cost, while the reversible mechanism remains a deterministic PEAK/ZERO substitution with an enumerative auxiliary stream. Gaussian-process threshold exploration and the Random Forest uniform-block agent are reported as diagnostic ablations; the deployed distance-one codec consists of the deterministic PEAK/ZERO mapping, CNN ranking, serialization-aware activation, and enumerative auxiliary stream.

The main scientific contributions of this paper are:

1. Serialization-aware net-payload-driven activation of reversible PEAK/ZERO mappings, where active tables are selected by useful payload after the exact

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serialized description cost required for deterministic recovery.

2. An executable enumerative wire representation for the active PEAK subset and the associated distance-one flip positions, yielding a decoder contract $(I', A_{\text{wire}}) \mapsto (I, M)$.
3. A learned candidate-cost ranker evaluated against Hamming-index, direct DRD-cost, transition-preserving, and connectivity-aware rankers, plus a CNN architecture ablation.
4. A common executable protocol comparing the proposed adaptive block-mapping method (ABM), the opposite-center-pair method (PPOCP), Huynh–Nguyen, and Dong with identical covers, messages, payload targets, reversibility checks, distance-reciprocal distortion (DRD) implementation, auxiliary-bit accounting, paired statistics, resources, and grouped steganalysis.

Section 2 positions the work in binary RDH and steganalysis. Section 3 defines the reversible mapping, learned ranking, and net-capacity codec. Section 4 describes the common protocol and reports document-image and BOSSbase results. Section 5 synthesizes the practical operating region, and Section 6 concludes the study.

2. Related Work

Binary-image RDH is constrained by the lack of grayscale redundancy: each pixel flip can break strokes, isolate components, or change local topology. The relevant literature therefore centers on pattern substitution, distortion control, and the side information required to restore the cover exactly.

2.1. Binary Pattern and Block Substitution

Early binary-image RDH uses logical operations and run-length or pattern statistics [13]. Ho et al. [14] established high-capacity reversible pattern substitution by exchanging frequent and eligible replacement patterns. Dong et al. [7] refine this line through adaptive overlapping patterns, PF/PFR handling, and a compressed location map. Yin et al. [9] select opposite-center pattern pairs (PPOCP), yielding a low-distortion profile. Zhang et al. [8] obtain very low distortion through image magnification, but the carrier dimensions change. Huynh and Nguyen [4] are the closest direct block-mapping reference: they use non-overlapping 3×3 blocks, frequent PEAK patterns, zero-frequency replacement patterns, and Hamming-distance-limited mappings. These works establish pattern substitution, 3×3 PEAK/ZERO block mapping, and Hamming-distance control as the relevant technical foundation.

Other binary-image variants expand the design space. Chhajed and Garg [3] use block-diagonal partition patterns, Ren et al. [6] use sliding windows, and Yang [5] prioritizes mixed black-white regions. They are useful contextual comparators, with executable wire models left open for a direct net-payload benchmark under the protocol used here.

2.2. Distortion, Coding, and Reconstruction Information

Binary document quality is measured with DRD because it accounts for spatially non-uniform perceptual sensitivity [15]. Feng, Lu, and Sun [12] further show that run continuity and connectivity are important when choosing flippable pixels. These results motivate the ranker ablations in Section 4.3: Hamming ordering alone is compared with direct DRD-cost, transition-preserving, connectivity-aware, and CNN-based rankings.

Coding-based RDH for binary covers also predates this work. Zhang et al. [16] improve binary-cover RDH schemes using optimal codes, and Zhang et al. [17] study optimal transition probabilities under general distortion metrics. Xiao et al. [18] propose a general-distortion RDH framework for binary covers using reconstruction information and matrix embedding; Li et al. [19] extend this line with multi-embedding. These studies establish distortion-aware binary-cover coding and reconstruction information as important conceptual prior art. Their contribution is complementary to the plaintext 3×3 PEAK/ZERO block-mapping codec developed here, where the serialized mapping description itself determines which reversible mappings remain active.

2.3. Encrypted, Halftone, and Learned Neighbors

Encrypted-domain binary RDH uses a different carrier model, but it is relevant to the treatment of side information. Ren, Lu, and Chen [11] use prediction in encrypted binary images, and recent encrypted-domain work includes adaptive block coding for grayscale carriers [20]. Halftone RDH also records dynamic embedding-state information [21]. These neighboring methods confirm that side information and compression are established tools; the distinction in this paper is their use as an exact serialized cost inside the active mapping-selection rule.

Learning serves here as candidate-cost ranking, while the reversible encoder remains deterministic and table based. Learned RDH and steganalysis show why such ranking and detection analyses are relevant [22–25]. Because PEAK/ZERO substitution changes local pattern histograms and transition rates, the experiments include grouped detectors to delimit the payload region where marked images become statistically separable from covers.

2.4. Research Gap

Table 1 consolidates the direct and conceptual neighbors. The relevant gap is an executable plaintext binary-image PEAK/ZERO method in which the active reversible mapping configuration is selected by useful payload after the exact serialized description cost needed for deterministic recovery. Within the reviewed corpus, this study contributes the joint treatment of serialization-aware activation, an enumerative PEAK/flip-position wire representation, and a common executable comparison against representative binary-image baselines.

Table 1
Comparative overview of binary RDH and related systems

Method (Year)	EC (bits)	Net accounting	PSNR (dB)	DRD	Adaptive	Uniform blocks	Steg. eval.
Ho et al. 2009 [14]	Published	Open	–	–	Yes	Pattern-dependent	Not extracted
Zhang et al. 2012 [16]	Code-dependent	Reconstruction info	–	Distortion bound	Yes	Binary cover sequence	Not extracted
Yin et al. 2020 [9]	1157	Open	27.17	0.23	Fixed	Reserved	Not extracted
Dong et al. 2020 [7]	∼1500	Open	∼25	<0.5	Yes	Reserved	Not extracted
Xiao et al. 2022 [18]	Variable	Reconstruction info	High	Optimized	Yes	Pixel sequence	Not extracted
Li et al. 2023 [19]	1000 example	Reconstruction info	49.45	–	Yes	Pixel sequence	Not extracted
Ren et al. 2023 [6]	∼2000	Open	∼24	<1	Fixed	Reserved	Not extracted
Zhang et al. 2019 [8]	369	Open	32.31	0.08	Fixed	Reserved	Not extracted
Chhajed & Garg 2023 [3]	<2000	Open	∼25	<1	Fixed	Reserved	Not extracted
Yang 2023 [5]	>2000	Open	∼23	<1	Partial	Reserved	Not extracted
Huynh & Nguyen 2025 [4]	2070	Open	24.13	0.55	Fixed	Reserved	Not extracted
Proposed	**2874 gross**	**2245 net**	**per-image**	**0.76**	**Yes**	**Evaluated**	**Yes**

EC: capacity reported by the cited study on its stated 512×512 protocol. Values provide cross-protocol context because datasets, payload definitions, and side-information accounting differ. "Steg. eval.": whether a steganalysis result was extracted for this overview table; Section 4.11 reports the common-protocol detector evaluation.

3. Net-Aware Learned Distance-One Block Mapping

3.1. Overview and Design Rationale

This section specifies the complete system evaluated in Section 4. The design uses the established 3×3 PEAK/ZERO block-mapping family, but the active mapping tables are chosen by useful payload after the exact serialized mapping description is charged. The encoder and decoder are therefore defined as the pair of maps

$$\mathrm{Enc}(I, M) \to (I', A_{\mathrm{wire}}) \tag{1}$$

and

$$\mathrm{Dec}(I', A_{\mathrm{wire}}) \to (I, M). \tag{2}$$

The auxiliary stream A_{wire} is stored or transmitted alongside the marked binary image I' as explicit synchronization metadata. The unconstrained adaptive threshold is retained as an ablation that characterizes the broader capacity–distortion landscape; the deployed codec uses globally disjoint Hamming-distance-one substitutions. Consequently, each active block carries one bit and changes at most one pixel.

The evaluated system proceeds in four deterministic steps:

- learns a perceptual ordering of absent distance-one patterns;
- enforces a globally injective PEAK/ZERO assignment;
- selects the frequency-ordered table prefix that maximizes net capacity; and
- serializes the payload length, active PEAK subset, and flip positions with an enumerative codec.

The embedding and extraction workflows are illustrated in Figures 1 and 2 and formalized in Algorithms 1 and 2.

3.2. Reversible Mapping Construction

Block representation and pattern histogram

Let I be a binary image of size $M \times N$. Complete non-overlapping 3×3 blocks are visited in raster order:

$$B_k = \{b_{i,j}\}, \quad i, j \in \{0, 1, 2\}, \quad k = 1, \dots, \left\lfloor \frac{M}{3} \right\rfloor \left\lfloor \frac{N}{3} \right\rfloor. \tag{3}$$

The resulting block count is the product of the two complete-block counts:

$$B = \left\lfloor \frac{M}{3} \right\rfloor \left\lfloor \frac{N}{3} \right\rfloor. \tag{4}$$

Each block is mapped to a decimal index:

$$d_k = \sum_{i=0}^{8} b_i 2^{8-i}, \tag{5}$$

where $b_i \in \{0, 1\}$ follows a fixed row-major ordering shared by both embedding and extraction; the first row-major bit is the most significant bit. Rows with indices at least $3\lfloor M/3 \rfloor$ and columns with indices at least $3\lfloor N/3 \rfloor$ are copied unchanged and never participate in capacity or extraction.

Uniform blocks ($d_k = 0$ and $d_k = 511$) are reserved for the separately evaluated Random Forest layer in Section 3.5; the core block-mapping histogram uses $d \in \{1, \dots, 510\}$. A histogram $H(d)$ is constructed for $d \in \{1, \dots, 510\}$.

Pattern sets

The implementation considers every non-uniform pattern occurring at least twice as an initial PEAK candidate. Candidates are sorted by decreasing frequency and then by numerical index. The pattern space is divided into:

- **PEAK candidates** $\mathcal{P}_0 = \{d : H(d) \geq 2, d \notin \{0, 511\}\}$,
- **Unused patterns (ZEROs)** $\mathcal{Z}$: patterns such that $H(d) = 0$,
- **Residual patterns**: patterns outside the embedding tables.

The final active set $\mathcal{P} \subseteq \mathcal{P}_0$ is selected by the explicit net-capacity rule defined below.

Pattern distance analysis

For each frequent pattern $P_i \in \mathcal{P}$ and each unused pattern $Z_j \in \mathcal{Z}$, the Hamming distance between their 3×3 representations is computed:

$$D_{i,j} = \sum_{p=1}^{9} \delta(P_i[p], Z_j[p]), \tag{6}$$

where $\delta(a, b) = 1$ if $a \neq b$ and 0 otherwise.

The set $\{D_{i,j}\}$ characterizes the distortion cost associated with mapping P_i to available unused patterns.

Adaptive candidate selection

Candidate patterns for each frequent pattern are selected adaptively. Let μ_i and σ_i denote the empirical mean and standard deviation of $\{D_{i,j}\}$. The adaptive Hamming threshold and candidate set are

$$H_{T,i} = \mu_i + \alpha\sigma_i \tag{7}$$

and

$$\mathcal{Z}_i = \{Z_j \in \mathcal{Z} \mid D_{i,j} \leq H_{T,i}\}, \tag{8}$$

where α controls the capacity–distortion trade-off.

This formulation relies solely on empirical statistics and remains distribution-free with respect to the distance values.

3.2.1. Globally Injective Mapping Construction

For the unconstrained ablation, let $k_i = |\mathcal{Z}_i|$ and define

$$L_i = \lfloor \log_2(k_i + 1) \rfloor \tag{9}$$

bits per occurrence. Exactly $2^{L_i} - 1$ candidates are retained:

$$\widetilde{\mathcal{Z}}_i \subseteq \mathcal{Z}_i, \qquad |\widetilde{\mathcal{Z}}_i| = 2^{L_i} - 1, \tag{10}$$

which makes the mapping

$$T_i : \{0,1\}^{L_i} \longleftrightarrow \{P_i\} \cup \widetilde{\mathcal{Z}}_i \tag{11}$$

a genuine bijection. After a ZERO is assigned, it is removed from the available pool; therefore the ranges of all T_i are globally disjoint.

The deployed distance-one policy restricts $L_i = 1$. Symbol 0 preserves P_i, while symbol 1 uses the lowest predicted-cost unassigned ZERO satisfying $D(P_i, Z) = 1$. Thus every modified block flips exactly one pixel.

Deterministic block traversal order

To ensure reproducibility and unambiguous extraction, a deterministic block traversal order is enforced.

The image is traversed in raster-scan order (left-to-right, top-to-bottom) at the block level. Each block is visited exactly once during both embedding and extraction.

During embedding, frequent patterns are processed in descending order of occurrence. However, the spatial traversal order of blocks remains unchanged. Each block is modified at most once, and block decisions remain independent.

Multi-pattern embedding strategy

Embedding scans spatial blocks once in raster order. When a block pattern is an active PEAK, the next bit selects either the unchanged PEAK or its assigned ZERO. The last symbol is truncated to the declared payload length during extraction; for the deployed one-bit codec, the payload boundary is unambiguous.

The total embedding capacity is:

$$C_{\text{gross}} = \sum_{P_i \in \mathcal{P}} n_i L_i, \tag{12}$$

where n_i denotes the number of occurrences of P_i.

3.3. Embedding Procedure

The embedding procedure operationalizes the mapping construction above as a deterministic encoder that constructs a single marked image. It first freezes the spatial support: complete non-overlapping 3×3 blocks are processed in raster order, while border pixels are copied unchanged. This convention is used for two reasons. First, it gives the decoder the same block sequence with coordinate-free synchronization. Second, it prevents boundary handling from becoming hidden auxiliary information that would distort the net payload comparison.

The next stage builds the reversible alphabet available in the current cover. Patterns with at least two occurrences are eligible PEAKs, absent patterns are ZEROs, and all-white/all-black uniform patterns are kept in the reserved uniform-block pool because changing them tends to create isolated specks or holes in document backgrounds and strokes. For each PEAK, admissible replacements are restricted to distance-one ZEROs. This restriction makes every embedded symbol a single-pixel block modification, keeps the restoration rule local, and allows the auxiliary stream to store only one base-9 flip position per active table. At this point, the CNN ranks admissible ZEROs by predicted perceptual cost; a global disjointness check then prevents two PEAKs from sharing the same stego pattern.

The final selection stage is the serialization-aware part of the method. After candidate assignments have been fixed, the encoder considers frequency-ordered prefixes of active mappings and selects the prefix that maximizes useful payload after charging the exact serialized description of that prefix. The message bits are then embedded during one raster scan: a payload bit 0 leaves an active PEAK block unchanged, while a payload bit 1 replaces it by the assigned ZERO. The declared message length q is serialized with the active table description, giving the decoder an explicit stopping point.

3.4. Efficient Extraction and Restoration

Extraction is designed as the exact inverse of the embedding contract, using the marked image and the serialized auxiliary stream. The auxiliary stream is decoded first because it fixes three pieces of synchronization information: the payload length, the active PEAK subset, and the single flip position that reconstructs each assigned ZERO. The decoder

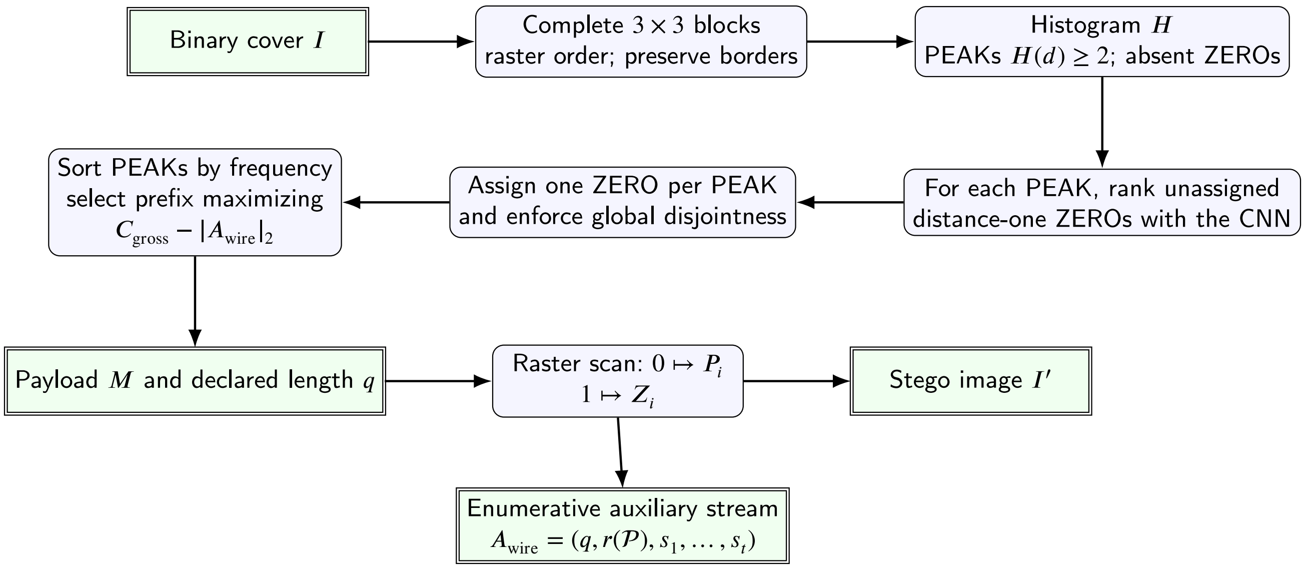

Figure 1: Evaluated embedding pipeline. Learned distance-one assignment is followed by explicit net-capacity optimization and enumerative serialization within a reversible-data-hiding synchronization model.

Algorithm 1 Net-Aware CNN-Ranked Distance-One Embedding

Require: Binary image I, payload $M = (m_1, \ldots, m_q)$, trained ranker f_θ
Ensure: Stego image I', serialized auxiliary stream A_{wire}

```
Enumerate complete 3 × 3 blocks and compute H(d)
P_0 ← {d : H(d) ≥ 2, d ∉ {0, 511}}; sort by (−H(d), d)
Z ← {d : H(d) = 0}; U ← ∅; T ← ∅
for each P_i ∈ P_0 do
    C_i ← {Z ∈ Z \ U : D(P_i, Z) = 1}
    if C_i ≠ ∅ then
        Z_i ← arg min_{Z∈C_i} (f_θ(I, P_i, Z), Z)
        T[P_i] ← {0 ↦ P_i, 1 ↦ Z_i}; U ← U ∪ {Z_i}
    end if
end for
Select the frequency-ordered prefix T* maximizing Σ_{P_i∈T*} H(P_i) − |A_wire(T*)|_2
if q > Σ_{P_i∈T*} H(P_i) then
    reject the requested payload
end if
I′ ← I; j ← 1
for each complete block of I in raster order do
    if its pattern is P_i ∈ T* and j ≤ q then
        replace it by T*[P_i][m_j]; j ← j + 1
    end if
end for
Serialize q, the active PEAK subset rank, and base-9 flip positions
return I′, A_wire
```

uses the serialized tables directly, making the transmitted synchronization contract the source of recovery decisions.

Once the active tables are reconstructed, they are collapsed into a single inverse dictionary. This is possible because the encoder enforces global injectivity: every stego pattern used by the codec belongs to at most one active mapping. Therefore, observing a PEAK pattern decodes bit 0 and restores the same PEAK, while observing its assigned ZERO decodes bit 1 and is also restored to the PEAK. Patterns outside the dictionary are copied unchanged. This dictionary view turns extraction into one linear raster scan with constant-time lookup.

The serialized payload length provides the stopping condition. After q bits have been recovered, remaining blocks are left unchanged because embedding used the same declared stopping point. If the scan finishes before q bits

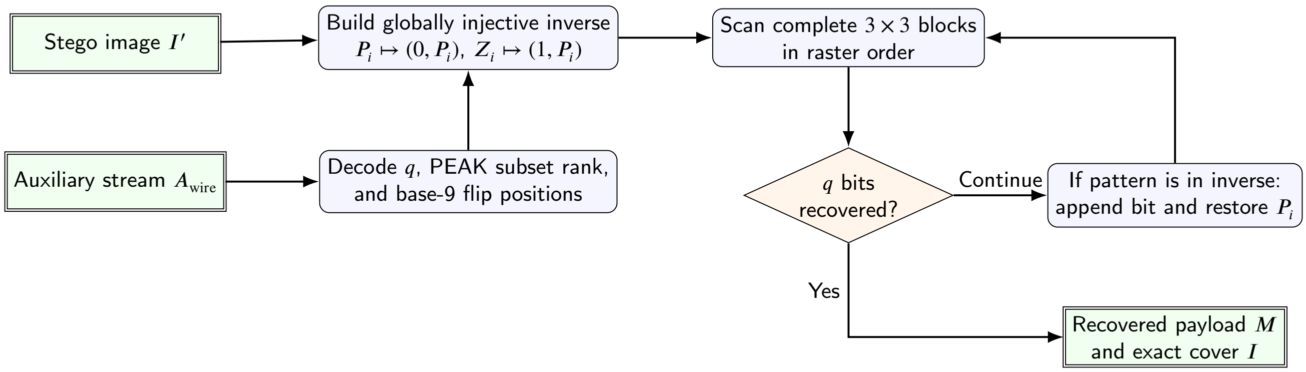

Figure 2: Extraction and restoration pipeline. The serialized payload length provides an unambiguous stopping condition, and incomplete border blocks remain untouched.

Algorithm 2 Deterministic Extraction and Exact Restoration

Require: Stego image I', serialized auxiliary stream A_{wire}
Ensure: Extracted message M, restored image I

Decode payload length q, active PEAKs, and their flip positions
Reconstruct all tables $T_i = \{0 \mapsto P_i, 1 \mapsto Z_i\}$
Build $\mathcal{D}[P_i] = (0, P_i)$ and $\mathcal{D}[Z_i] = (1, P_i)$
$I \leftarrow I'$; $M \leftarrow \emptyset$
for each complete block of I' in raster order **do**
    if $|M| = q$ **then**
        break
    end if
    Compute pattern index d
    if $d \in \mathcal{D}$ **then**
        Append the decoded bit to M and replace the block by its PEAK
    end if
end for
if $|M| \neq q$ **then**
    reject the inconsistent stego/auxiliary pair
end if
return M, I

are recovered, the pair (I', A_{wire}) is rejected as inconsistent. This explicit rejection rule is necessary because exact reversibility is guaranteed for matched marked-image and auxiliary-stream pairs transmitted over a lossless channel.

To reduce extraction complexity, all mapping tables are merged into a single inverse dictionary:

$$\mathcal{D} : p \mapsto (m, P_i), \tag{13}$$

where p is a stego pattern, $m \in \{0, 1\}$ the extracted bit, and P_i the original pattern.

3.5. Learned Ranking, Ablations, and Wire Accounting

Distortion characteristics

For each embedded block mapped from P_i to $\mathcal{Z}_i$, the number of flipped pixels is bounded by:

$$\max_{Z_j \in \mathcal{Z}_i} D_{i,j}, \tag{14}$$

ensuring pattern-specific distortion control.

Gaussian-process diagnostic

The parameter α is explored by Gaussian-process Bayesian optimization. For every evaluated value, the implementation records capacity, PSNR, DRD, and the normalized objective

$$J(\alpha) = \lambda \left(1 - \frac{C_{\mathrm{total}}(\alpha)}{C_{\mathrm{ref}}}\right) + (1 - \lambda)\mathrm{DRD}(\alpha). \tag{15}$$

Expected Improvement selects the next value of α [26]. The equal weighting $\lambda = 0.5$ gives the two normalized diagnostic terms identical influence; it serves only the exploratory scalarization of the unconstrained diagnostic. A λ sweep therefore characterizes the exploration stage, whereas all final tables use the independently specified distance-one policy. Optimizing α alone provides incomplete control of DRD < 1, motivating the constrained distance-one policy used by the integrated system. The exploratory search uses $\alpha \in [0, 1.5]$, initial points $\{0, 0.5, 1, 1.5\}$, six EI iterations on

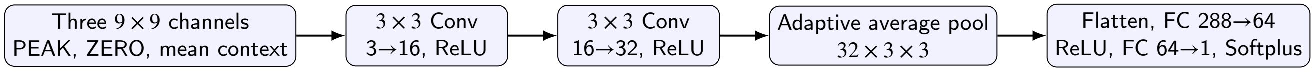

Figure 3: Candidate-cost CNN used to rank globally available distance-one ZERO patterns. The scalar output estimates reciprocal-distance replacement cost.

a 151-point grid, $\lambda = 0.5$, exploration 0.01, and a Matérn-5/2 kernel with a white-noise term. The random seed is 20260607.

CNN-based candidate ranking

For a PEAK pattern and each distance-one ZERO candidate, the CNN receives three 9×9 channels: the centered PEAK, the centered candidate, and the average spatial context of all PEAK occurrences. It regresses the empirical reciprocal-distance replacement cost. The candidate with the smallest predicted cost is assigned to symbol 1, while symbol 0 preserves the PEAK.

The 9×9 support covers the central 3×3 block and a three-pixel margin in every direction. It contains the complete 5×5 DRD support around each of the nine possible flipped pixels, including pixels outside the embedding block; a 7×7 patch covers corner flips only partially. Larger patches add context beyond the local training target.

The network contains two 3×3 convolutional layers with 16 and 32 channels and ReLU activations, adaptive 3×3 average pooling, and a $288 \rightarrow 64 \rightarrow 1$ regressor with ReLU and Softplus. Training uses Smooth-L1 loss, AdamW with learning rate 10^{-3}, batch size 128, 20 epochs, and seed 20260607. Complete source images define the train/validation split at image level.

Figure 3 summarizes the evaluated architecture and its scalar candidate-cost output.

Random-Forest uniform-block ablation

Uniform blocks are processed by a second reversible layer. A Random Forest classifier predicts whether a block admits a low-cost single-pixel flip from its 9×9 context, and a second forest predicts the pixel position. The safety label uses a local reciprocal-distance cost at most 0.5. Selected coordinates and positions are compactly serialized for exact extraction and restoration. The experiment shows that the classifier identifies reliable locations, while the coordinate stream defines the next compression target. The final net-aware configuration therefore assigns its bit budget to the enumerative block-mapping layer.

Both forests use 120 trees and maximum depth 14. The safety model uses minimum leaf size 2 and balanced class weights; the position model uses minimum leaf size 1. Validation is grouped by source image with three folds, seed 20260607, and a deployment probability threshold of 0.7. These bounded defaults were fixed before the net-capacity evaluation: 120 trees stabilize the grouped predictions, depth 14 limits memorization of image-specific contexts, and the 0.7 threshold favors precision when every accepted block incurs coordinate overhead. They were fixed before evaluating the reported test outcomes. This layer is an evaluated ablation, and the final payload figures focus on the enumerative block-mapping layer because the current coordinate stream outweighs the measured gross gain.

Net-capacity accounting

Useful capacity subtracts the exact serialized side information:

$$C_{\text{net}} = C_{\text{gross}} - |A_{\text{wire}}|_2. \tag{16}$$

For distance-one mappings, the wire format ranks the sorted active-PEAK subset in the combinatorial number system. The corresponding nine possible flip positions are then represented as base-9 digits. This representation closely matches the information content of the mapping set and is more compact than independent fixed-width pairs. Mapping pairs are ordered by occurrence count, and the net-maximizing prefix is retained. When every subset has nonpositive useful capacity, the encoder returns the unchanged image.

For t active PEAKs $0 \le p_1 < \cdots < p_t < 512$, the subset rank is

$$r(\mathcal{P}) = \sum_{j=1}^{t} \binom{p_j}{j}. \tag{17}$$

If $s_j \in \{0, \ldots, 8\}$ is the flipped bit position, the combined integer is

$$u = r(\mathcal{P}) 9^t + \sum_{j=1}^{t} s_j 9^{t-j}. \tag{18}$$

For example, with active PEAKs $\{5, 9, 12\}$ and flip positions $(0, 4, 8)$, $r = \binom{5}{1} + \binom{9}{2} + \binom{12}{3} = 261$, the base-9 component is 44, and the combined state is $u = 261 \times 9^3 + 44 = 190313$. The wire stream contains a one-byte raw/compressed marker, a four-byte magic identifier, a one-byte format version, the 32-bit big-endian payload length q, the 16-bit big-endian table count t, and the shortest big-endian byte string capable of representing the joint state space $\binom{512}{t} 9^t$. Thus the variable state is charged as

$$8 \left\lceil \frac{\left\lceil \log_2 \left(\binom{512}{t} 9^t \right) \right\rceil}{8} \right\rceil \tag{19}$$

bits, and the fixed ABM header is 96 bits. The active PEAKs are sorted by increasing pattern index inside the combinatorial rank, while the frequency-ordered prefix determines which PEAKs are active. Image dimensions are provided by the stego container and are checked by the extraction

Table 2
Executable auxiliary-information model used for exact extraction and restoration.

Method	Auxiliary element	Specified in cited paper	Executable representation	Charged bits
ABM	Encoding marker	Proposed codec	raw/compressed marker byte	8
ABM	Stream identity	Proposed codec	magic and version bytes	40
ABM	Payload length q	Proposed codec	big-endian uint32	32
ABM	Active table count t	Proposed codec	big-endian uint16	16
ABM	Active PEAKs and flips	Proposed codec	byte-aligned joint state in Eq. 19	variable
Huynh–Nguyen	Peak, selected states, q	Partial	compact byte stream with zlib option	measured
Dong	Context divisor, end position, q, PF/PFR map	Partial	packed and zlib-compressed location map	measured
PPOCP	Pair level, used centers, original centers, q	Partial	varint positions plus bit-packed original centers	measured

Image dimensions are supplied by the marked-image container for ABM and checked by the decoder API. Baseline synchronization formats are treated as conservative executable formats for the common round-trip protocol.

API. The stream is a serialized plaintext synchronization channel; confidentiality and authentication, when required by an application, are orthogonal transport-layer services.

Table 2 gives the executable synchronization model used in the comparisons. For ABM the listed representation is part of the proposed codec. For baselines, the original publications specify the reversible principle and some parameters while leaving the complete independent byte stream open; the table therefore records the conservative executable representations implemented for this protocol.

3.6. Comparison with Conventional Block-Mapping

Huynh–Nguyen [4] is the closest direct 3×3 PEAK/ZERO block-mapping comparator. The comparison therefore isolates one difference: the block representation, the PEAK/ZERO idea, and Hamming-distance control are established foundations, while the evaluated change is serialization-aware activation. The same reversible mapping family is retained when its exact encoded description leaves useful payload after Equation 16. On the eight document benchmarks, the proposed implementation provides 2874 gross bits and 2245 net bits on average after 629 auxiliary bits are charged. Huynh–Nguyen report 2070 gross bits under their own protocol; this published value is contextual because their serialized auxiliary cost remains open.

3.7. Computational Complexity and Formal Properties

The following propositions summarize the asymptotic cost and recovery guarantees of the deployed embedding and extraction algorithms. The analysis uses the finite 512-state 3×3 alphabet to show that histogram, mapping, and inverse-dictionary structures remain constant with respect to image size, while image traversal dominates runtime. The final propositions establish deterministic reversibility and block independence.

Notation. Let $n = MN$ denote the total number of pixels and $B = \lfloor M/3 \rfloor \lfloor N/3 \rfloor$ the number of complete 3×3 blocks. Let $\mathcal{P}$ be the set of frequent (PEAK) patterns and $\mathcal{Z}$ the set of unused (ZERO) patterns. For a pattern $P_i \in \mathcal{P}$, let n_i be the number of blocks exhibiting P_i, L_i the embedding capacity associated with P_i, and $E = \sum_i n_i^{\mathrm{emb}}$ the total number of blocks actually used for embedding.

Proposition 3.1 (Embedding Time Complexity)**.** *The time complexity of the deployed distance-one embedding algorithm is*

$$O\big(n + |\mathcal{P}_0|\, c_\theta + E\big) = O(n), \tag{20}$$

where c_θ is the fixed cost of evaluating at most nine candidates with the fixed CNN.

Proof. The algorithm first partitions the image into blocks and computes pattern indices, which requires a single pass over all pixels and thus $O(n)$ time. Histogram construction and pattern classification are performed in constant time per block, yielding $O(B) = O(n)$ complexity. For each PEAK, its at most nine distance-one neighbors are generated directly and ranked by the fixed-size CNN, requiring $O(|\mathcal{P}_0|c_\theta)$ time. Since the binary 3×3 pattern space contains only 512 states, this term is bounded independently of image size. Finally, each embedding operation consists of a constant-time table lookup and block substitution, and is executed for E blocks, contributing $O(E)$ time. Summing all terms gives the stated complexity. □

Proposition 3.2 (Embedding Space Complexity)**.** *The space complexity of the embedding algorithm is*

$$O(n + q). \tag{21}$$

Proof. The algorithm stores the original and modified images, requiring $O(n)$ space. The histogram, mapping tables, and inverse dictionary contain at most 512 states and therefore require constant space with respect to image size. The payload buffer requires $O(q)$ space. □

Proposition 3.3 (Optimized Extraction Time Complexity)**.** *The optimized extraction algorithm runs in*

$$O(n) \tag{22}$$

time.

Proof. Extraction scans the stego image once to reconstruct blocks and compute pattern indices, which requires $O(n)$ time. The inverse dictionary has at most 512 entries. Each complete block is processed at most once in deterministic raster order with constant-time lookup, and extraction stops after the declared q bits. □

Proposition 3.4 (Extraction Space Complexity)**.** *The space complexity of the optimized extraction algorithm is*

$$O(n + q). \tag{23}$$

Proof. During extraction, memory is required for the stego image, the recovered image, and the inverse dictionary constructed from the mapping tables. The inverse dictionary is bounded by the 512-state pattern alphabet, and the message buffer stores exactly q recovered bits. □

Proposition 3.5 (Deterministic Reversibility)**.** *Given an unmodified stego image and its valid auxiliary stream, the proposed embedding and extraction procedures recover the payload and cover exactly.*

Proof. Each table maps 0 to its PEAK and 1 to one absent distance-one pattern. Assigned ZEROs are globally disjoint and were absent from the cover, so the union of table ranges is injective. During extraction, the inverse dictionary therefore maps every active stego pattern to one bit and one original PEAK. Since blocks are non-overlapping, modified at most once, and processed in a deterministic traversal order, the declared payload length gives a unique stopping point. Unused complete blocks and incomplete border pixels are never modified. Hence both the embedded message and original image are reconstructed exactly. □

Proposition 3.6 (Independent Blocks)**.** *Embedding operations on distinct blocks are independent.*

Proof. Each block is processed independently using a pattern-specific mapping table. Each operation is confined to the current block and uses the block's own state. Therefore, embedding decisions are local and preserve inter-block independence. □

4. Experimental Evaluation

4.1. Datasets, Splits, and Reproducibility

The evaluation has two complementary scales. The eight standard 512 × 512 binary document images used by Huynh–Nguyen [4] support direct validation against the block-mapping literature; their diversity is shown in Figure 4. BOSSbase [27] provides 10,000 grayscale images, from which seed 20260608 selects 100 profile images and 100 disjoint test images. The profile split supplies the image-level statistics needed by the conservative Dong implementation. Test images are binarized at threshold 128. The CNN ranker trained on the document images is transferred directly to BOSSbase, preventing test-set adaptation.

The four-method protocol embeds identical pseudorandom messages at targets 64, 128, 256, and 512 bits. A run counts as available after exact message extraction and bitwise cover restoration. Huynh–Nguyen uses the published threshold $T = 5$. Dong evaluates all context divisors $l \in \{1, \dots, 10\}$ and retains the exact minimum-DRD result. PPOCP and Dong use conservative serialized synchronization or location information where the publications leave the independent wire representation open. Accordingly, net-payload conclusions apply to the executable wire formats defined in this study, while DRD and availability provide a comparison less sensitive to byte-stream conventions.

4.2. Metrics and Statistical Protocol

Gross payload is the message length, auxiliary bits are the exact serialized wire length, and useful payload is $C_{\text{net}} = C_{\text{gross}} - C_{\text{aux}}$, where $C_{\text{aux}} = |A_{\text{wire}}|_2$. Distortion is measured with changed pixels, PSNR, and DRD [15]. Availability is the fraction of test images that carry the target and complete the exact round trip.

Primary per-payload comparisons use common available images. A secondary fixed cohort contains the 46 images on which all four methods are available at all four payloads. ABM is paired with each baseline for net bits and DRD using the Wilcoxon signed-rank test and Holm correction within each payload. Mean net payload is accompanied by a 95% percentile bootstrap interval with 10,000 image-level resamples and seed 20260608. Runtime includes embedding, serialization, extraction, and verification. Memory is peak process RSS minus pre-call RSS, sampled every 2 ms. Experiments used Python 3.13.5, NumPy 2.2.6, PyTorch 2.8.0 CPU, scikit-learn 1.7.2, and SciPy 1.17.1 on 64-bit Windows with 12 logical processors.

4.3. Serialization-Aware Activation Ablation

Table 3 isolates the central algorithmic decision. All variants use the same CNN distance-one candidate assignment; only active table selection and exact wire charging differ. A0 reports gross payload before wire cost and is therefore a gross-reporting diagnostic reference point. A1 charges the wire after activating all feasible tables. A2 is the proposed serialization-aware activation, which selects the frequency-ordered prefix that maximizes Equation 16. Relative to A1, A2 reduces auxiliary information by 180 bits on average and raises mean net payload by 105.9 bits (4.95%) while preserving exact reversibility on all eight document images.

4.4. Candidate Ranking and CNN Architecture

Table 4 compares the learned ranker with simple non-learning alternatives under the same admissible distance-one ZERO set and common payload per image. The CNN is close to the direct DRD-cost ranker and improves mean DRD by about 5% relative to Hamming-index ordering. The reversible mechanism remains the deterministic, globally injective PEAK/ZERO substitution defined in Section 3.

Table 5 gives the controlled CNN architecture diagnostic. All configurations use 20 epochs, AdamW with learning rate 10^{-3}, Smooth-L1 loss, batch size 128, the same two-image train/validation split, and deterministic seeds. The deployed two-layer 9×9 model is retained as a compact compromise: it is close to the lowest-MAE wider model, faster at inference, and keeps held-out placement DRD within the same practical range.

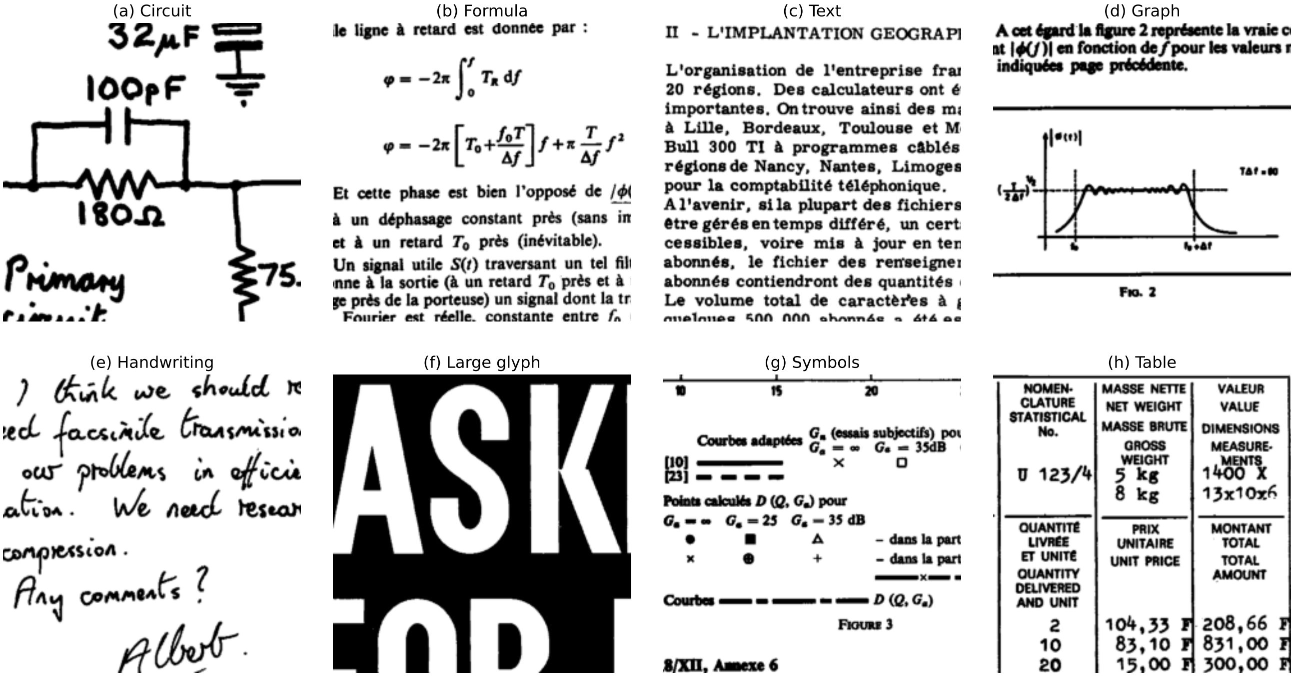

Figure 4: Eight binary document images used for block-mapping validation.

Table 3
Central ablation of serialization-aware mapping activation on the eight document images. Values are means in bits except DRD; the post-hoc value in A0 is the net payload obtained when the exact wire is charged after the gross-only selection.

Variant	Active-table selection	Wire charged	Gross cap.	Payload	Aux.	Net payload	Mean DRD
A0	all feasible; gross reported	Gross only	3932	2948	0	2948 (2139 post hoc)	0.771
A1	all feasible; wire charged	Charged	3932	2948	809	2139	0.778
A2 proposed	serialized-net prefix	Charged	3832	2874	629	2245	0.761

Table 4
Ranker ablation with identical distance-one candidates and common payloads.

Ranker	Mean net	Mean DRD	Round trip
Hamming index	2141	0.822	Exact
Direct DRD cost	2139	0.778	Exact
Transition preserving	2140	0.829	Exact
Connectivity aware	2140	0.820	Exact
CNN	2137	0.780	Exact

4.5. Serialization Sensitivity for Baselines

The original descriptions of PPOCP, Dong, and Huynh–Nguyen specify the reversible principles while leaving the complete standalone byte stream partly unspecified. The common protocol therefore uses conservative executable synchronization. We also ran a header-credit sensitivity check: for each baseline, the method-identifying header, version field, and image-shape fields are removed from the charged cost, giving a 104-bit credit while retaining the payload-critical maps, states, locations, and payload length. With this credited accounting, ABM retains the highest mean net payload on every all-method matched cohort: −57 vs. −120 bits at 64, −7 vs. −72 bits at 128, 93 vs. 23 bits at 256, and 309 vs. 200 bits at 512 for the best credited baseline. This fixed-header variation leaves the useful-capacity ordering unchanged for the tested executable representations.

4.6. Machine-Learning Validation

Image-wise leave-one-out (LOSO) validation trains eight independent CNNs, each on seven document images and tests it on the remaining complete image. Across 13,808 held-out candidates, pooled mean absolute error is 0.0367 and pooled Pearson correlation is 0.9893; per-image correlations range from 0.9861 to 0.9939. A separate LOSO placement experiment applies each fold model to its held-out image at an identical 75% distance-one payload. Mean DRD decreases from 0.8054 with deterministic Hamming-index ordering to 0.7663 with CNN ordering, a 4.85% reduction. All eight images improve, with per-image reductions from 2.83% to 7.94%, while every message and cover is recovered exactly. The separate two-image architecture diagnostic in Table 5 evaluates 4323 held-out candidates under the fixed 20-epoch protocol. Thus, regression accuracy and

Table 5
CNN architecture diagnostic on the two-image detailed split. Inference time is measured on 4323 held-out candidates. Held-out DRD is the mean DRD after model-ranked placement on the diagnostic held-out images. All configurations share the reversible mapping and auxiliary stream.

Configuration	Params	Train s	Inf. μs	MAE	Pearson	Held-out DRD
1 conv., 9×9, w24	14625	6.97	7.18	0.0716	0.962	0.848
2 conv., 9×9, w24	38865	23.62	27.19	0.0339	0.991	0.830
3 conv., 9×9, w24	44073	27.80	38.21	0.0347	0.991	0.848
2 conv., 7×7, w24	38865	18.11	19.51	0.0383	0.989	0.845
2 conv., 11×11, w24	38865	27.77	56.13	0.0485	0.985	0.825
2 conv., 9×9, w16	23649	15.16	24.86	0.0379	0.990	0.839
2 conv., 9×9, w32	56385	27.27	40.62	0.0303	0.992	0.847

Table 6
Net-aware results on the eight binary document images.

Image	Gross	Aux.	Net	DRD
circuit-2	1576	592	984	0.583
formula-5	3692	672	3020	0.900
graph1-5	1978	632	1346	0.662
handwr2-8	2351	632	1719	0.768
large-8	1138	488	650	0.616
symbol-6	2367	672	1695	0.897
table1-3	4161	656	3505	0.789
french-4	5729	688	5041	0.864
Average	**2874**	**629**	**2245**	**0.760**

operational DRD impact are evaluated separately, with fold-level regression and placement results provided with the reproducibility artifacts.

The uniform-block Random Forest is evaluated by source-image-grouped cross-validation on 14,798 samples. It reaches ROC AUC 0.995, precision 0.897, recall 0.970, and flip-position accuracy 0.971. In the combined reversible pipeline, the layer adds 181.4 gross bits per document image on average, while its current coordinate representation uses 2691.0 auxiliary bits. The resulting combined net payload is therefore −264.6 bits on average, compared with 2245.0 bits for the base enumerative block-mapping layer. This identifies coordinate coding as a concrete optimization opportunity and keeps the Random Forest layer as a diagnostic component.

4.7. Document-Image and External Net Capacity

Table 6 reports the integrated CNN-ranked mapping layer at a 75% payload budget. Auxiliary sizes are measured from serialized bytes, and all eight messages and covers are recovered exactly.

On the 100 BOSSbase test images, net-aware table selection gives 669.6 gross, 377.1 auxiliary, and 292.5 net bits on average. Median net capacity is 159 bits, 75 images are net positive, mean DRD is 0.300, and every evaluated message and cover is recovered exactly. Disabling net-aware table selection on the same sample yields 832.5 gross, 731.4 auxiliary, and 101.1 net bits on average, showing that the active-prefix decision also improves the external transfer setting.

4.8. Multi-Payload Common-Protocol Results

Table 7 and Figure 5 summarize the all-method matched subsets. ABM moves from metadata-dominated operation at 64 bits to a median-positive point at 128 bits and a clearly positive operating region at 256 and 512 bits. PPOCP defines the lowest-DRD curve, while ABM defines the highest-net curve.

Table 8 preserves the ordering observed in the payload-specific matched analysis. It also removes changing-cohort composition as an explanation for the positive ABM net payload from 128 bits onward.

Figure 6 complements the table with availability and the net-bits–DRD operating curves. PPOCP carries every target on all test images. ABM availability progresses from 89% at 64 bits to 70% at 512 bits, matching Dong at the highest target and remaining above Huynh–Nguyen. The right panel is descriptive because the methods span different DRD ranges; Table 8 provides the controlled cross-payload comparison.

4.9. Paired Significance and Binarization Strata

At each payload, ABM's paired net advantage over Dong, Huynh–Nguyen, and PPOCP remains significant after Holm correction. At 256 bits, adjusted p-values are below 6.7×10^{-13}, 3.1×10^{-13}, and 2.2×10^{-14}, respectively. Median ABM net gains are 160, 1752, and 3016 bits, with paired rank-biserial correlations of 0.974, 1.000, and 1.000. DRD tests place ABM below Dong and above the perceptually focused PPOCP and Huynh–Nguyen curves, yielding a two-objective interpretation.

Binarization suitability is defined before embedding: threshold 128 must agree with Otsu on at least 95% of pixels, and foreground fraction must lie in $[0.05, 0.95]$. Twenty-two test images satisfy this criterion. As shown in Figure 7, ABM is available on all 22 through 256 bits, while net-positive behavior at 256 and 512 bits is present in both strata. Suitability therefore serves as an applicability indicator, with net performance reported separately in both strata.

At 512 bits, the 70 available images have mean foreground fraction 0.289, 189.2 eligible non-uniform PEAK

Table 7
Mean net bits and median DRD on all-method matched images.

		Proposed ABM		Dong		Huynh–Nguyen		PPOCP	
Target	**Matched**	**Net**	**DRD**	**Net**	**DRD**	**Net**	**DRD**	**Net**	**DRD**
64	86	−57	0.041	−224	0.056	−2200	0.035	−997	0.023
128	83	−7	0.080	−176	0.108	−2096	0.070	−1653	0.041
256	69	**93**	0.135	−81	0.200	−1812	0.117	−2856	0.064
512	46	**309**	0.176	96	0.332	−1191	0.178	−4822	0.097

Table 8
Fixed-cohort sensitivity analysis on the 46 images available for every method and payload. Values are mean net bits with 95% bootstrap intervals; DRD is the cohort median.

Target	Method	Mean net bits (95% CI)	Median DRD
64	Proposed ABM	−48.5 [−49.0, −48.0]	0.022
	Dong / Huynh–Nguyen / PPOCP	−226.3 [−229.6, −223.0] / −1644.7 [−1854.3, −1447.5] / −1028.0 [−1063.3, −990.1]	0.029 / 0.024 / 0.012
128	Proposed ABM	8.0 [5.7, 10.1]	0.043
	Dong / Huynh–Nguyen / PPOCP	−178.3 [−187.8, −169.9] / −1582.8 [−1792.4, −1386.8] / −1707.8 [−1767.5, −1643.0]	0.066 / 0.044 / 0.025
256	Proposed ABM	113.9 [108.3, 119.0]	0.087
	Dong / Huynh–Nguyen / PPOCP	−83.8 [−105.6, −65.7] / −1456.3 [−1656.2, −1260.9] / −2954.1 [−3060.2, −2835.8]	0.132 / 0.084 / 0.050
512	Proposed ABM	309.0 [289.7, 325.9]	0.176
	Dong / Huynh–Nguyen / PPOCP	95.8 [55.1, 131.0] / −1190.6 [−1397.6, −996.7] / −4821.7 [−5025.6, −4607.1]	0.332 / 0.178 / 0.097

patterns, and non-uniform 8×8 block fraction 0.227. The 30 unavailable images have corresponding means 0.025, 34.2, and 0.024. Unavailable cases are therefore concentrated in near-empty or nearly uniform thresholded photographs with too few repeated non-uniform patterns; available cases still restore the message and cover exactly.

4.10. Runtime and Memory

Figure 8 and Table 9 report the 256-bit resource benchmark on ten protocol images. PPOCP provides the shortest runtime, Huynh–Nguyen the smallest incremental memory profile, and ABM requires a median runtime of 3.56 s and median incremental peak RSS of 2.56 MiB. Dong's complete ten-divisor optimization quantifies the computational cost of per-image context search.

Table 10 decomposes the proposed method on the eight document images at the 75% payload budget. Candidate generation, including CNN-ranked distance-one assignment, and embedding dominate runtime; the enumerative serialization itself is negligible.

4.11. Grouped Steganalysis

The security evaluation uses foreground fraction, horizontal and vertical transition rates, and normalized 2×2 and 3×3 pattern histograms. Five-fold GroupKFold keeps

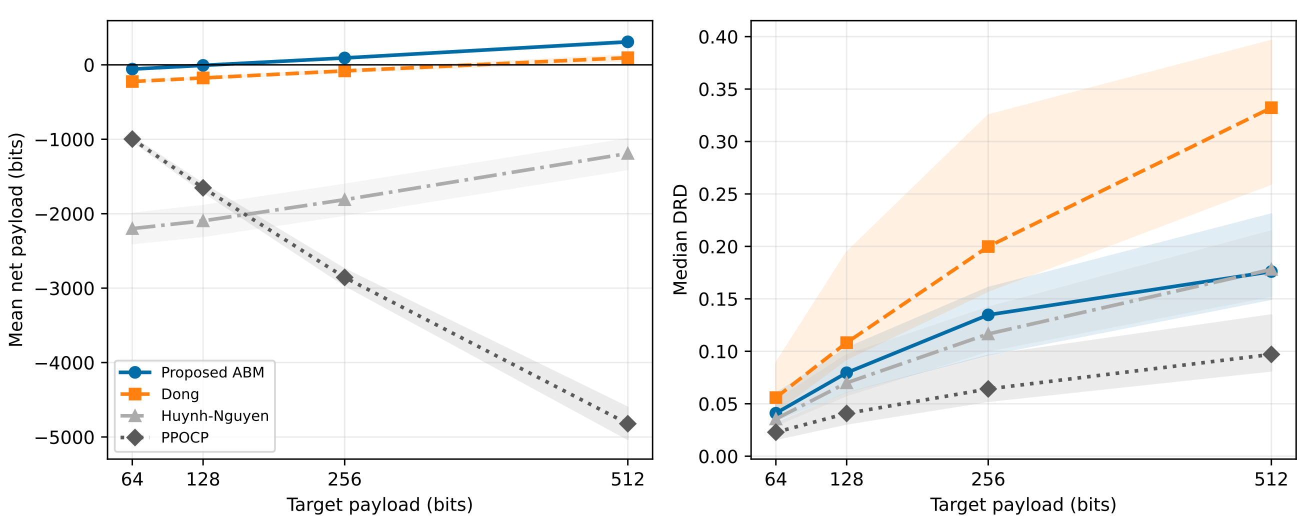

Figure 5: Payload scaling on all-method matched images: mean net payload (left) and median DRD (right), with 95% bootstrap confidence bands.

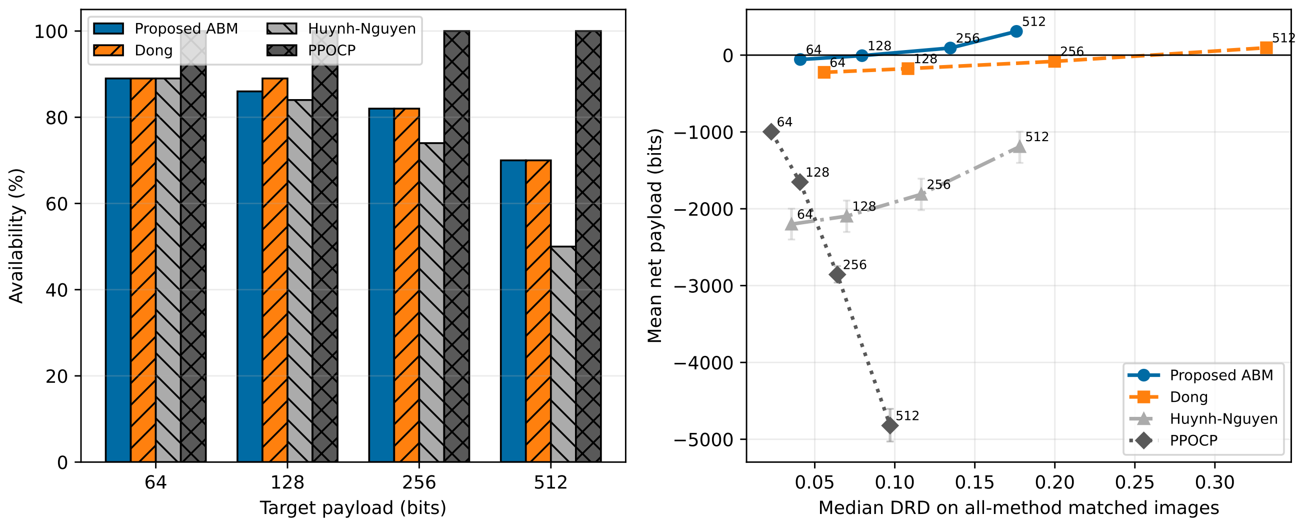

Figure 6: Complementary views of payload feasibility and useful-capacity efficiency. Error bars denote 95% bootstrap confidence intervals for mean net payload; numbers on the right panel denote target payloads in bits.

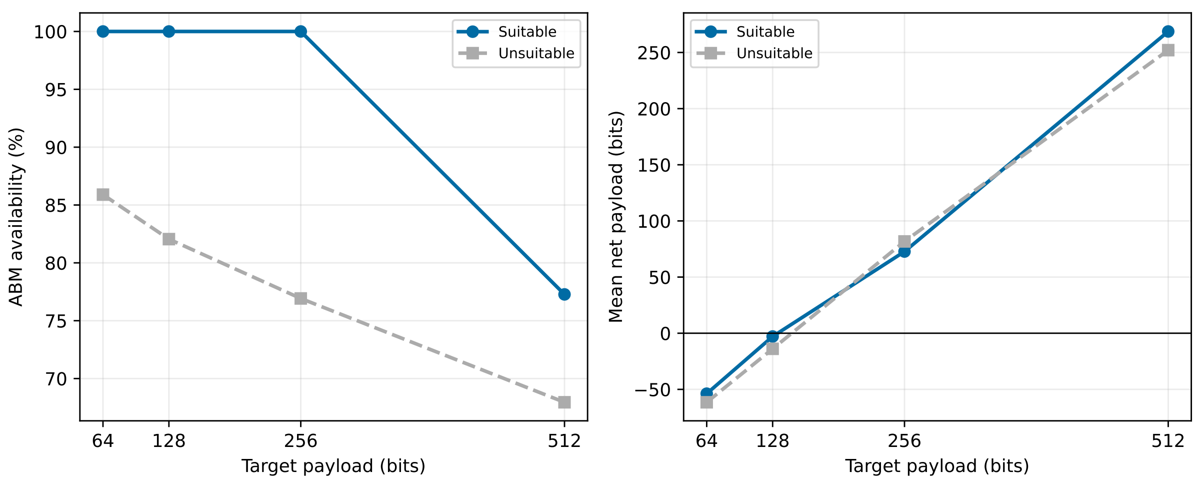

Figure 7: ABM availability and mean net payload after stratification by fixed-threshold/Otsu agreement.

each cover and its stego in the same fold. Both a standardized logistic model and a 500-tree Random Forest are evaluated.

Figure 9 visualizes the complete AUC curves, and Table 11 reports the logistic detector. PPOCP provides the most resistant curve from 128 to 512 bits. ABM's AUC increases with useful payload, making statistical footprint the principal signal for future optimization. The Random Forest gives ABM AUC values 0.623, 0.691, 0.769, and 0.851, confirming the same payload trend with a different model.

A document-image payload sweep explains this trend. When the ABM payload fraction increases from 25% to 100%, mean net payload rises from 329 to 3203 bits, but the mean Jensen–Shannon divergence between cover and marked 3×3 pattern histograms also rises from 0.0087

Table 9
Median resource profile at a 256-bit target.

Method	Available	Time (s)	Peak RSS increase (MiB)
Proposed ABM	9/10	3.56	2.56
Dong	9/10	16.97	0.05
Huynh–Nguyen	7/10	1.51	0.02
PPOCP	10/10	0.16	0.70

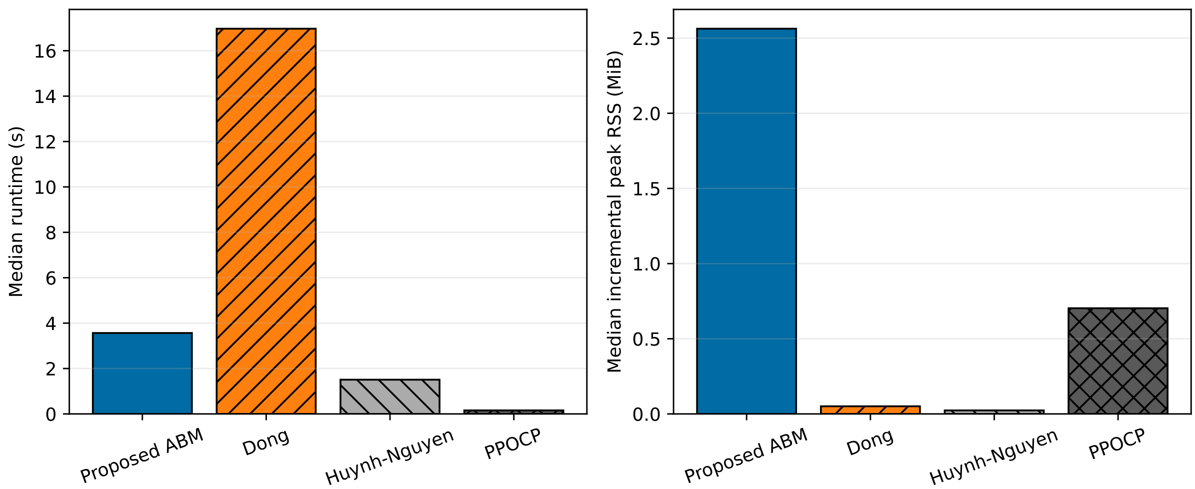

Figure 8: Median wall-clock runtime and incremental peak RSS at 256 bits.

Table 10
ABM runtime decomposition by module on document images.

Module	Median time (s)
Candidate generation and ranking	0.5771
Serialized-prefix optimization	0.2832
Embedding	0.7155
Serialization	0.00015
Extraction	0.1926
Round-trip verification	0.00008
Total	1.7857

to 0.0409. Mean horizontal and vertical transition-rate displacements rise from 0.0032/0.0032 to 0.0127/0.0128. The steganalysis features therefore respond to the same PEAK/ZERO redistribution that produces useful capacity: repeated PEAK patterns are depleted and distance-one ZERO patterns are populated. High payload is therefore best interpreted as capacity-oriented reversible annotation in access-controlled channels. Figure 10 gives the qualitative counterpart: residual locations show how each method distributes its changes under the same 512-bit target on table1-3.

5. Discussion

5.1. Positioning Across Practical Objectives

The experiments separate four objectives that are frequently merged in RDH comparisons. In useful capacity, ABM provides the highest net-payload curve and crosses into mean-positive net payload between 128 and 256 bits. In perceptual quality, PPOCP defines the lowest median DRD, followed closely by Huynh–Nguyen at the higher target. In applicability, PPOCP has universal target availability, while ABM combines high availability with compact side information. In computational cost, ABM remains practical and is substantially faster than the complete Dong parameter search.

This decomposition supports a specific, protocol-bounded position: among the four evaluated implementations, ABM provides the highest net payload after serialized metadata, whereas PPOCP provides the lowest-distortion and most favorable detectability profile. This conclusion rests on exact round trips, multi-payload curves, a fixed-cohort sensitivity analysis, and paired tests. Because synchronization formats materially affect net payload, the result is framed as a protocol-bounded comparison of the evaluated implementations.

5.2. Security-Aware Optimization Direction

The measured AUC curves and content-adaptive detectability principles [28] suggest a direct route for strengthening the method. The current CNN predicts reciprocal-distance cost, whereas the steganalysis features respond to shifts in local pattern distributions. A security-aware ranker can optimize

$$\mathcal{L}(Z_j) = \beta\,\widehat{\mathrm{DRD}}(Z_j) + \gamma\,\Delta_{\mathrm{hist}}(Z_j) + \eta\,\Delta_{\mathrm{trans}}(Z_j), \quad (24)$$

where Δ_{hist} measures the change in local pattern histograms and Δ_{trans} measures transition-rate displacement. The coefficients can be selected on a Pareto front of net bits, DRD, and grouped detector AUC. This extension builds directly on the validated ranker and preserves the exact reversible wire format.

The present measurements define the operating envelope. At 256 and 512 bits, logistic AUC values of 0.900 and 0.957 place the current ABM configuration in a capacity-oriented reversible-annotation regime, despite its positive net payload and exact reversibility. At 64 bits, AUC 0.706 reflects lower detectability, although metadata overhead dominates useful capacity. PPOCP remains the preferred evaluated method for security-prioritized operation, whereas ABM addresses applications in which useful reversible capacity is primary and the stego channel itself is access-controlled.

The RF uniform-block experiment points to a second optimization path: coordinate coding can use enumerative

Table 11
Grouped logistic-detector ROC AUC on matched cover/stego pairs.

Target	Pairs	Proposed ABM	Dong	Huynh–Nguyen	PPOCP
64	86	0.706	0.656	**0.599**	0.653
128	83	0.826	0.759	0.688	**0.654**
256	69	0.900	0.784	0.774	**0.691**
512	46	0.957	0.896	0.909	**0.690**

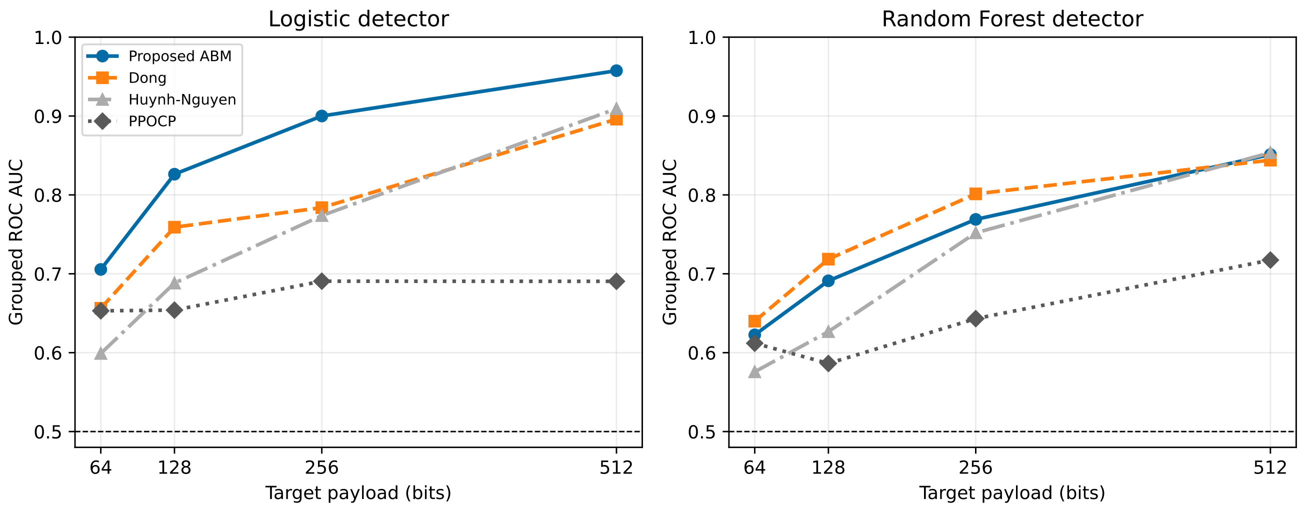

Figure 9: Grouped steganalysis across payloads with logistic (left) and Random Forest (right) detectors. The dashed line denotes chance-level AUC.

subset coding conditioned on the deterministic block order. Its grouped classification metrics show that location selection is reliable; compressing the selected subset targets the measured source of overhead.

5.3. Experimental Scope

The eight document images provide continuity with the binary block-mapping literature, and the disjoint BOSSbase protocol adds 100 external test images, four payloads, paired inference, resources, and steganalysis. BOSSbase binarization should be read as a transfer test, with a large scanned document corpus remaining a natural next benchmark. The 512-bit all-method cohort contains 46 images, so both availability and fixed-cohort results are reported. The logistic AUC of 0.957 at that payload identifies a high-order statistical footprint even though the visual DRD remains moderate. These measured boundaries define the validated operating region and provide explicit targets for distribution-aware ranking. The reversible channel assumes lossless storage and

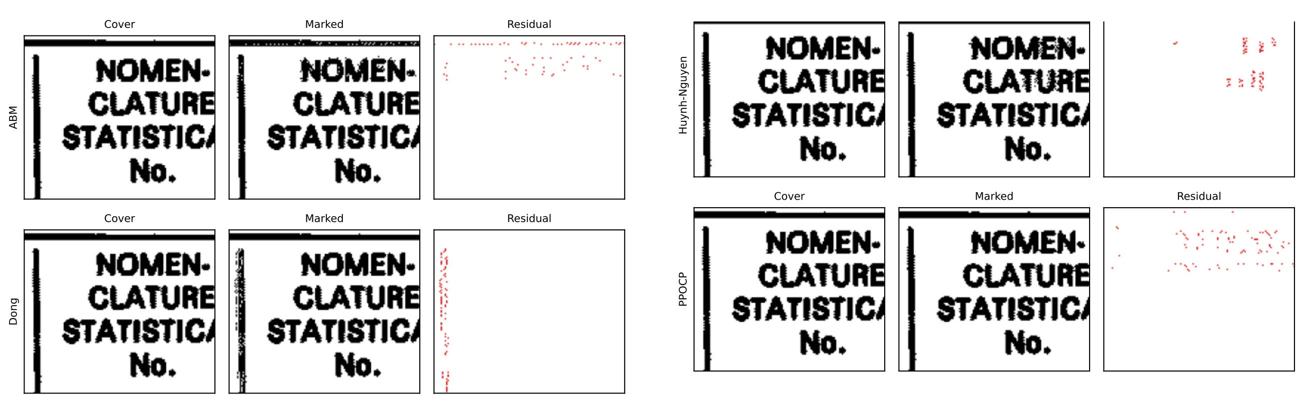

Figure 10: Cover, marked, and residual zooms on table1-3 at a 512-bit target, split into two panels to fit the two-column layout: ABM and Dong appear on the left, and Huynh–Nguyen and PPOCP on the right. Residual pixels are shown in red on the marked crop to make modification locations visible under the same payload.

Table 12
Channel stress test for ABM with the same auxiliary stream.

Channel	Stego	Message	Cover
PNG	Yes	Yes	Yes
TIFF	Yes	Yes	Yes
JPEG q=5	Altered	Mismatch	Mismatch
One-pixel noise	Altered	Mismatch	Mismatch

transmission. A stress test on table1-3 at 256 bits confirms exact recovery after PNG and TIFF round trips, while JPEG recompression at quality 5 and one-pixel perturbation convert the input into mismatched stego/auxiliary pairs, illustrating the required lossless-channel assumption (Table 12). Geometric, noisy, or lossy transformations therefore require an external synchronization or error-control layer.

6. Conclusion

This work develops and evaluates serialization-aware activation for net-payload-driven PEAK/ZERO block mapping in binary-image RDH. The enumerative codec makes the active mapping description executable and charged; the CNN ranker improves candidate placement on held-out images as a secondary cost-ordering component. The complete system delivers 2245 net bits on average on eight document images and 292.5 net bits on 100 BOSSbase images. Every reported available experiment restores the message and cover exactly.

Under the common protocol, ABM obtains mean net payloads of −57, −7, 93, and 309 bits at targets 64, 128, 256, and 512 bits on matched images. Its paired net advantage is significant against every baseline after Holm correction; at 256 bits, all adjusted p-values are below 6.7×10^{-13}. Together, availability, binarization, runtime, memory, and two-classifier steganalysis make this result a reproducible practical profile beyond a single capacity number.

Under the stated implementations and wire-level accounting, the combined evidence identifies ABM as the highest useful-capacity curve in the evaluated common protocol, while PPOCP retains the lowest measured distortion and most favorable detectability profile. This evidence therefore adds a concrete net-capacity benchmark and an exactly reversible reference implementation. The measured security curves also identify local-distribution preservation as the main optimization target while retaining compact metadata and exact restoration.

Acknowledgements

This work was supported by the authors' institutions.

CRediT authorship contribution statement

Stéphane Gaël R. Ekodeck: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Vivien Beyala Kamgang:** Conceptualization, Methodology, Writing – review and editing. **Serge Alain Ebélé:** Conceptualization, Methodology, Validation, Writing – review and editing. **Julius Marcellin Nkenlifack:** Supervision, Project administration, Writing – review and editing.

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